Alfonso David Balladares Quintana

Date of birth: 26/03/2001 | **Place of birth:** Lima, Peru | **Nationality:** Peruvian | **Gender:** Male | **Phone number:** (+51) 961287969 (Mobile) | **Email address:** alfo.david26@gmail.com | **Website:** linkedin.com/in/alfonso-balladares | **Address:** 15067, Lima, Peru (Home)

ABOUT ME

Bachelor of pharmacy and biochemistry with a research background in bioprospecting, metabolomics, natural products characterization and formulation of drug delivery systems. Practical experience includes operating high-resolution analytical platforms (UHPLC-HRMS-QTOF) and the use of specialized software for bioinformatics. Co-author of peer-reviewed publications in natural products chemistry and tradional medicine. I am particullary interested in omic sciences and drug discovery.

WORK EXPERIENCE

RESEARCHER – UNIVERSITY ANTONIO RUIZ DE MONTOYA – 06/2026 – Current – LIMA, PERU

Project Title: *Development of a biopesticide via nanoencapsulation of coffee and cocoa subproducts: A sustainable circular economy alternative to enhance coffee value-chain productivity under climate change.*

- **Biopolymer Extraction & Purification:** Execute and compare extraction protocols to isolate pectin and polysaccharide biopolymers from agricultural subproducts (coffee, cocoa, and grape waste) targeting high yield and purity.
- **Chemical characterization and biological activities:** Perform and validate diverse in vitro spectrophotometric assays (e.g., DPPH, ABTS, FRAP) to quantify antioxidant activity, total phenolic and flavonoid content in extracted natural fractions.
- **Phytochemical Valorization:** Evaluate the pcycle agricultural waste into high-value bioactive raw materials for sustainable delivery systems.

LABORATORY ASSISTANT – CENTER FOR COSMETIC RESEARCH, TECHNOLOGY, AND INNOVATION (CITIC) - UNIVERSITY SAN IGNACIO DE LOYOLA – 01/2026 – 08/2026 – LIMA, PERU

- Design and formulate advanced topical delivery systems, including emulsion, anhydrous matrices and specialized transdermal patches incorporating nanofibers.
- Optimize the incorporation of natural bioactive ingredients sourced from the Peruvian Amazon into stable cosmetic vehicles, ensuring physical-chemical stability and target compound integrity.
- Co-draft technical project proposals and experimental designs for national funding grants (CONCYTEC / PROCIENCIA) and collaborative R&D bids with private pharmaceutical and cosmetic enterprises.

LABORATORY ASSISTANT – BIOTECHNOLOGY AND OMICS IN LIFE SCIENCES (BIOLIFS) - NATIONAL UNIVERSITY OF SAN MARCOS – 05/2024 – 12/2025 – LIMA, PERU

- Conducted metabolomic profiling and compound screening of endophytic bacteria (Rahnella aceris, *Pseudomonas lurida*, and *Pantoea agglomerans*) isolated from high-altitude Andean medicinal plants
- Operated UHPLC-MS/MS systems and utilized advanced metabolomics software (MetaboScape) to identify and annotate complex small molecules, including phytohormones, phenolics, cordycepins, and pesticides
- Overseen the preparation of critical reagents, culture media, and buffers, while managing the operational maintenance of laboratory equipments.

QUALITY INTERN – BOEHRINGER-INGELHEIM – 07/2024 – 07/2025 – LIMA, PERU

- Conducted rigorous technical data audits and organoleptic quality evaluations to support the compliant batch release of human and veterinary pharmaceuticals under global GxP standards.
- Investigated out-of-specification (OOS) laboratory results, performing root-cause analyses to generate Corrective and Preventive Actions (CAPAs) and design robust Quality Agreements.
- Maintained precise electronic batch records and technical documentation utilizing specialized quality management systems (Veeva Vault, Trackwise, SAP) to ensure total data traceability.

PHARMACY INTERN IN TRADITIONAL AND COMPLEMENTARY MEDICINE – ESSALUD (SOCIAL HEALTH INSURANCE OF PERU)-INTERNSHIP – 12/2019 – 03/2020 – LIMA, PERU

- Researched traditional medicine practices and native medicinal flora to evaluate their chemical profiles and potential integration into evidence-based public health protocols.
- Assisted in the design, pilot compounding, and quality control of traditional phytotherapeutic preparations and galenical formulations under standardized laboratory conditions.
- Compiled literature reviews and technical documentation to support clinical and pharmacological research in complementary and integrative medicine.

RESEARCH ASSISTANT – REPRONAT – NATIONAL UNIVERSITY OF SAN MARCOS – 10/2019 – 07/2025 – LIMA, PERU

- Applied analytical methods including Thin-Layer Chromatography (TLC), Column Chromatography, and UV-Vis Spectrophotometry for the separation and structural characterization of organic compounds.
- Contributed to the preparation and publication of research papers on natural product characterization.
- Standarized protocols for extraction, fractionation, and chromatographic separation of secondary metabolites—specifically flavonoids—from native medicinal plants.

EDUCATION AND TRAINING

02/2018 – 02/2024 Lima, Peru
BACHELOR OF PHARMACY AND BIOCHEMISTRY National University of San Marcos

LANGUAGE SKILLS

Mother tongue(s): **SPANISH**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	B2	B2	B2	B1	B1
FRENCH	B2	B2	B1	B1	B1

PUBLICATIONS

2024
Caracterización y potencial antimicrobiano de Luma chequen (Molina) A. Gray [Flavonoid Characterization and Antimicrobial Activity of the Methanolic Extract of Luma chequen (Molina) A. Gray Leaves]

Authors: Fernández-Rebaza G, Balladares-Quintana A, Soto-Tantaraico E, Zeña-Chiara E, Salazar-Salvatierra M, Bonilla-R

2023
Leche de cabra (Capra aegagrus hircus): una revisión de su valor nutricional, actividades biológicas, tratamiento térmico y productos alimenticios derivados [Goat Milk (Capra aegagrus hircus): A Review of its Nutritional Value, Biological Activities, Thermal Treatment, and Derived Food Products].

Authors: Zeña Chiara EN, Huamanchay Meza AV, Balladares Quintana AD, Guadalupe-Sifuentes de Posadas LF, Arias Arroyo GC

2021
Composición química y caracterización de flavonoides de extractos metanólicos de hojas de dos tipos de Ruta chalepensis L. [Chemical Composition and Flavonoid Characterization of Methanolic Leaf Extracts of Two Types of Ruta chalepensis L.]

Authors: Bonilla Rivera PE, Toche Tuesta A, Fernández Rebaza GA, Curioso Melo DE, Rayme Gutiérrez RL, Soto Tantaraico EC, Balladares Quintana AD, et al

2021

Alimentos con potencial efecto inmunomodulador y antiviral a propósito de la pandemia COVID-19 [Foods with Potential Immunomodulatory and Antiviral Effects in the Context of the COVID-19 Pandemic]

Authors: Huaccho-Rojas J, Yánac-Tellería W, Balladares Quintana A, et al.

2020

Revisión del efecto antiviral e inmunomodulador de plantas Medicinales a propósito de la pandemia COVID-19 [Review of the Antiviral and Immunomodulatory Effect of Medicinal Plants in the Context of the COVID-19 Pandemic]

Authors: Huaccho-Rojas J, Balladares Quintana A, Yánac-Tellería W, Rodríguez CL, Villar-López M..

CONFERENCES AND SEMINARS

22/06/2026

Acquisition of a UPLC System Coupled with a High-Resolution Mass Spectrometer (QTOF) to Strengthen the Foodomics Research Line: "Role and function of microbial and plant bioactive compounds from native Peruvian products on the Gut Microbiota"

CONFERENCES, WORKSHOPS & PRESENTATIONS
Speaker & Technical Demonstrator | *National University of San Marcos (UNMSM) / PROCIENCIA*

- Presented original metabolomic and bioprospecting research results obtained through the high-resolution UPLC-QTOF platform, demonstrating the identification of active microbial metabolites.
- Co-led the practical workshop sessions, demonstrating sample preparation, chromatographic optimization, and high-resolution mass spectrometry (QTOF) data acquisition workflows to academic and industry participants.

SKILLS

Scientific Writing | Reading Comprehension | Patent Drafting & Review | perform scientific research | Chromatographic separation (TLC, column, flash, steric exclusion, ion-exchange resin)

CERTIFICATIONS

World Intellectual Property Organization (WIPO / OMPI), 05/2026

Intellectual Property & Patent Search Course

Key Competencies: Patent search methodologies, prior art analysis, patent drafting structures, and IP protection strategies for biotechs and natural products.

Mode of learning: Online

Universidad Peruana Cayetano Heredia (UPCH), 10/2025

Scientific Paper Writing

Completed a highly selective 32-hour (2 ECTS credits equivalent) professional development program through the Postgraduate School, focusing on manuscript structuring, peer-review strategies, and academic publishing standards in biomedical sciences.

Mode of learning: Online